ENAE 694
Spacecraft Communications



Class 1 • The Spacecraft Communications Problem

Dave Israel • August 31, 2026

Tonight Has Three Parts

Part 1 — Welcome, syllabus, and introductions

Part 2 — Communications as a mission system

Part 3 — Lunar orbiter: volume, rate, latency, and architecture

Introductions

Dave Israel

disrael@umd.edu

(443)895-1864

Not Simply Radio. Not Simply Networking.

This course is about moving mission information through a complete system.

The Semester Leads to a Complete Network

COURSE GRADE • Homework & applied work 70% • Final project 30%			
MEETING	**CLASS FOCUS**	**MEETING**	**CLASS FOCUS**
Aug. 31	Mission system	**Nov. 2**	Demodulation, BER & coding
Sept. 14	Received power & geometry	**Nov. 9**	Multiple access, crosslinks & relays
Sept. 28	Antennas & pointing	**Nov. 16**	Coverage & availability
Oct. 5	Optical communications	**Nov. 23**	Data services & end-to-end flow
Oct. 19	Noise, G/T & complete link budget	**Nov. 30**	Routing, store-and-forward & DTN
Oct. 26	Signals, modulation & bandwidth	**Dec. 7**	Interoperable networks + architecture studio
NO CLASS • Sept. 7 — Labor Day • Sept. 21 — Yom Kippur • Oct. 12 — Fall Break			

What You Will Be Able to Do

Translate mission needs and provider capabilities into communications design targets

Build and interpret complete link budgets

Analyze coverage, availability, capacity, and latency

Trace data from its source to its final user

Compare architectures and defend the trades

Two Perspectives, One Complete System

Mission view: what information must arrive—and when?

↓

Provider view: what services can shared assets deliver?

↓

Radio links

+

Networks and operations

↓

Mission information reaches its user

We Will Do the Engineering in the Room

Introduce the principle

Work an example together

Change one assumption

Predict what changes—and why

Use homework to extend the example

The Communications Problem

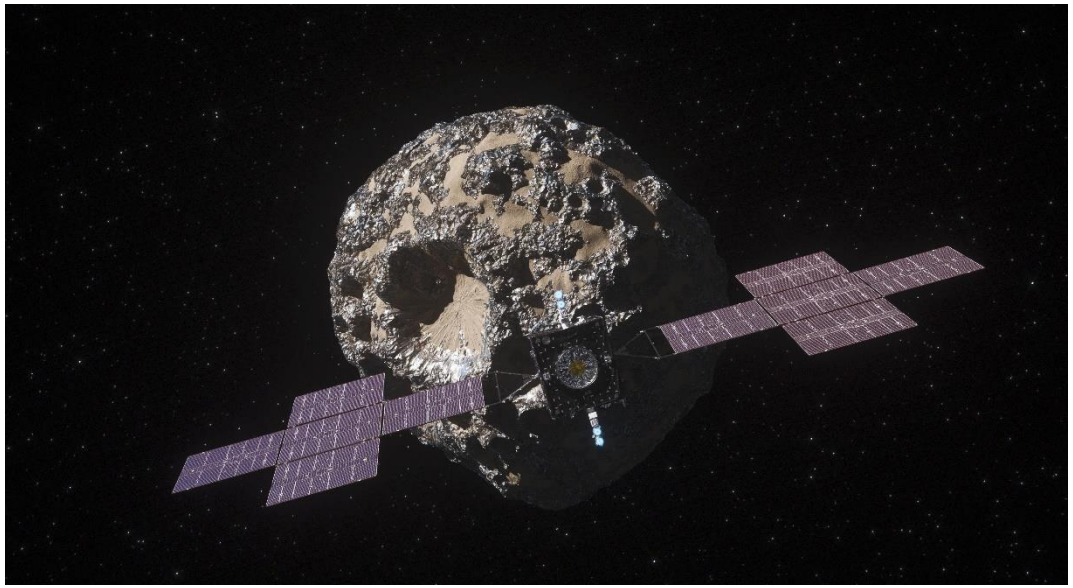














Mission Needs Come First

“Use Ka-band” is an implementation choice.

“Use a relay” is an architecture choice.

Start with what information must reach whom, how much, how quickly, and how dependably.

Then compare ways to make that happen.

Start With Eight Questions

1. Where is the information produced?
2. Where must it ultimately go?
3. How much must be delivered?
4. How soon must it arrive?
5. How complete and reliable must delivery be?
6. Are some data more important than others?
7. What security is required?
8. What resources and constraints limit the design?

Start With Source and Destination

Geometry sets range, visibility, and propagation time

Interfaces identify the systems and organizations involved

The final destination may be far beyond the ground antenna

Different sources may need different paths and services

Data Volume Is Not a Data Rate

A volume becomes a rate only after we know how much communication time is available.

First Worked Example: A Lunar Orbiter

8 GB of science data each day

Two scheduled 10-minute Earth contacts per day

Assume every scheduled second can carry science data

What is the minimum data rate?



First Convert the Daily Volume

$$8 \text{ GB/day} \times 8 \text{ bits/byte}$$

$$= 64 \text{ Gbit/day}$$

Keep the units visible. They are often the first line of defense against nonsense.

Then Divide by Available Contact Time

Contact time = 2 × 10 min × 60 s/min = 1,200 s

Minimum rate = 64×10^9 bits ÷ 1,200 s

≈ 53.3 Mbit/s

53.3 Mbit/s Is Only a Lower Bound

Protocol and coding overhead

Acquisition and handover time

Changing geometry and link performance

Lost or shortened contacts

Telemetry, commands, and competing users

Variation in data production

Now Add a Latency Need

The orbiter also produces an event alert.

Very small data volume

Ten-minute delivery deadline

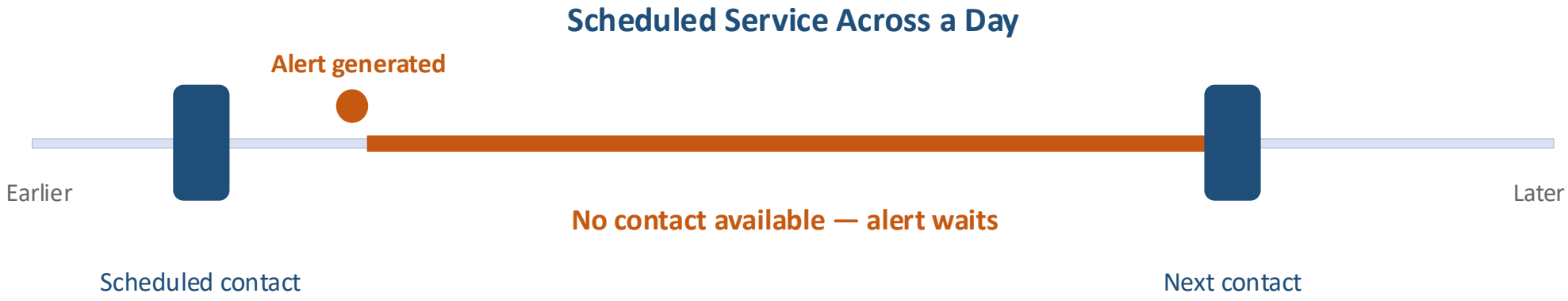
Does the twice-daily contact schedule satisfy the mission?

Capacity and Availability Are Different

Capacity asks: how much?

Availability asks: when?

Enough total capacity does not guarantee a usable path at the critical moment.



Different Data Need Different Services

Science imagery — high volume, hours of latency, very high completeness

Event alert — tiny volume, 10-minute latency, urgent

Housekeeping — moderate volume, priority changes with spacecraft state

Command — tiny volume, mission-dependent latency, authorized and correct

The smallest data product may drive the most expensive design choice.

Three Plausible Architectures



Architecture Discussion

Which option meets the mission’s alert deadline?

Which uses shared capacity well across missions?

Which creates the greatest provider dependence?

Which offers the greatest resilience?

What do the mission and provider still need to know?

Not Enough Information Is a Result

Name the uncertainty.

Identify the decision it affects.

Ask for the evidence that would resolve it.

Recognizing that we cannot choose yet is engineering judgment.

Constraints Are Part of the Design

Size, weight, and power

Spectrum and interference

Cost, schedule, complexity, and risk

Shared provider capacity and scheduling

Everything is a trade. The equations help us make the trade honestly.

Spectrum and Security Shape Design

Spectrum: allocation, bandwidth, and interference

Commands: authorization and integrity

Telemetry and science: confidentiality may differ

Networks: every provider interface creates a trust boundary

Resilience and security are related—but they are not the same thing

A Design Must Work End to End

Link budget — can a representative link work?

Coverage — is the service available when needed?

Data delivery — can information arrive in time?

Mission use — does the result support operations?

Assumptions — what could change the answer?

A Working Vocabulary

Link — one transmitter-to-receiver connection

Service — a capability delivered to a user

Provider — shared assets and operations serving multiple missions

Network — connected assets and rules that move data end to end

Architecture — the chosen arrangement of functions, interfaces, and resources

Six Questions Test the Complete System

1. **Does the link close?**
2. **Is service available when needed?**
3. **Can the network carry the offered load?**
4. **Does the data reach the final user on time?**
5. **Can the provider serve this mission along with the others?**
6. **Can we defend the complete architecture?**

Before You Leave

Name one mission communications need

for a mission you know

— and —

one fact still missing

before you could design the system.

Homework 1 Extends Tonight's Example

1 — Daily data volume

2 — Contact time and ideal payload rate

3 — Capacity and timeliness

Due September 14 before class • Show calculations and units • Revise until correct